

NAGESH KURE

AI/ML Engineer — Agentic AI & Backend Systems

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in Agentic AI and backend systems — LLM orchestration, multi-agent workflows (LangChain/LangGraph), MCP-based tool integration, and geospatial AI. Built MAPCHAT at Avineon India Ltd, a production agentic platform combining a multi-agent conversational system with an OCR/computer-vision document-intelligence pipeline. Improved response accuracy from 80% to 90% and cut LLM token consumption ~50%. Strong FastAPI/Python backend foundation with experience across RAG, vector databases, LLM observability, and scalable API architecture.

TECHNICAL SKILLS

Agentic AI / GenAI: LangGraph, LangChain, MCP, Multi-Agent Systems, RAG, LLMs, Prompt Engineering, LLM Evaluation

Backend & APIs: Python, FastAPI, REST APIs, SQLAlchemy ORM, Pydantic, JWT Auth, Redis

Computer Vision / OCR: PaddleOCR, YOLO (v8/v11/v26), Tesseract, EasyOCR, Object Detection

Databases: PostgreSQL, MySQL, MongoDB, FAISS, Pinecone, ChromaDB, Vector Databases

Infra / Tools: Langfuse, Docker, Git, Postman, Kafka, Azure, PyTorch, TensorFlow, Hugging Face

WORK EXPERIENCE

AI/ML Engineer — Backend — Avineon India Ltd, Hyderabad

Present

Project: MAPCHAT — AI-Powered Geospatial Intelligence Platform

- Built the backend for MAPCHAT, an agentic geospatial platform with a multi-agent conversational router (LangChain, LangGraph, MCP) and a PDF/OCR document-intelligence router — using FastAPI, Redis, and ArcGIS.
- Designed multi-agent orchestration with Human-in-the-Loop (HITL) validation for natural-language geospatial queries; integrated MCP to replace Flask-based tool calling, improving execution efficiency.
- Reduced LLM token consumption by ~50% and improved response accuracy from 80% to 90% via agent orchestration and prompt optimization on GPT-OSS 120B; integrated Langfuse for LLM observability.
- Built a PDF/OCR extraction pipeline (PaddleOCR, YOLOv26) converting scanned property croquis into structured JSON, achieving 90% OCR accuracy and 95% object-detection accuracy; added confidence scoring and packaged with Nuitka/PyInstaller.

AI Engineer Intern — Codetru, Hyderabad

Jun 2025 — Present

Project: Japfu.AI

- Built an AI-powered onboarding/document automation system (PAN, Aadhaar, Passports) reducing verification time by 40% and achieving 95% extraction accuracy via OCR.
- Implemented MCP for tool integration across onboarding, authentication, and HR approval workflows; added role-based auth with MySQL.

Backend Developer Intern — Codetru, Hyderabad

Apr 2025 — Jun 2025

Project: Ruto.ai

- Built RESTful APIs with FastAPI, PostgreSQL, and SQLAlchemy ORM, improving request processing speed by 25%; added Pydantic validation (~30% input errors) and JWT auth.

AI/ML Intern — Konu, Hyderabad

Feb 2025 — Apr 2025

Project: Real Estate Assistant

- Built a multi-agent real estate chatbot (property search, financing, EMI) achieving 90% response accuracy; migrated tool integration to MCP, improving stability by 15%.

ADDITIONAL PROJECTS

- ANPR System — Real-time vehicle number-plate detection (YOLOv8) at 92% accuracy; Flask/FastAPI interface with MySQL storage.
- Loan Prediction System — Logistic Regression model (85% accuracy) with Streamlit web app deployed on Render.

EDUCATION

Master of Computer Applications (MCA) — Pratibha Institute of Business Management, Pune

2022 — 2024

B.Sc, Computer Science — MGM Dr. GY Pathrikar College of CS & IT, Aurangabad

2018 — 2021

CERTIFICATIONS

- Data Science and AI — Under the mentorship of Omkar Nallagoni, Senior Data Scientist, Hyderabad